

DESIGN AND DEVELOPMENT OF AUTOMATIC SOLAR PANEL CLEANING SYSTEM BASED ON ARDUINO

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ABSTRACT

This project is about the design and development of a solar panel cleaning system. The main object of this design prototype is to clean the solar panel using an electrical mechanism, such that efficiency and quality of solar panel is not compromised. As a matter of fact, Gulf region - especially Saudi Arabia - is facing a lot of dust storms and the solar panels need to be cleaned frequently. If task is performed manually, it will be very costly and time consuming. Water sprinklers and a special wiping material shall be used in the conceived mechanism design to insure quality of cleaning. Solar panel is vulnerable to accumulated dust on its surface. The efficiency of the solar panel gradually decreases because of dust accumulation. Two-step mechanism used in this system consists of an exhaust fan which works as an air blower and a wiper to swipe the dust from the panel surface. A dc motor is used to power the wiper. Since, the system does not need water to clean solar panel, it avoids the wastage of water and effective cleaning in desert areas.

Keywords: Arduino, Dust Removal, Sand Removal, Solar Panel Cleaner.

I. INTRODUCTION

To ensure sustainable environment, solar energy can play a vital role because it is an enormous, inexhaustible and green source of energy. Solar energy is directly usable for electrical power generation for a variety of applications which includes residential, commercial and industrial uses. It can provide the necessary amount of electricity without any threat to the environmental and health issues because no harmful gasses are emitted at the time of energy conversion. Moreover, the total sunlight that hits the surface of the earth in an hour and a half, according to the US department of energy, is adequate to manage the yearly energy consumption of the whole world. It is estimated that, the total amount of fossil fuel which is stored in the earth is equivalent to the energy produced from the sunshine on earth of around 18 days. In a solar energy system, photovoltaic (PV) solar panel provides DC electricity from the continuous flow of energy from the sun. MPPT method based on Fuzzy logic controller is presented into produce maximum energy from PV system. Once the installment of solar panels has been completed, the fuel is free. The operating cost of the system is very low when compared to other energy production methods. However, solar energy is CO₂ free renewable energy source, the routine maintenance of solar panel is required. The particles of dust on the solar panel come mainly from urban and industrial products. SiO₂, Al₂O₃, Fe₂O₃, CaMg(CO₃)₂, Ca(OH)₂, CaO and CaCO₃ are some sorts of dust particles found on the solar panel. Dust Accumulation on the surface of solar panel has serious impact on the system's efficiency. It is estimated that, about 50% of system's efficiency can be reduced and almost 15% power losses can be occurred in dry areas. Therefore, it is essential to keep solar panel's surface clean as much as possible. Solar energy is the cleanest, inexhaustible, easily available and an abundant energy resource available on the earth. Solar energy can suffice the present and future energy demands of the world. The energy industry is also keen on moving towards providing the electricity in green and sustainable manner. Despite the huge initial cost for deployment, photovoltaic systems are being used widely to meet the growing electricity demand. The lifetime of Photovoltaic panel is around 30 years [1]. Photovoltaic systems are being used widely to fulfill the growing demand for electricity. Despite, the huge cost of initial deployment, the solar panel installation is increasing day by day. Photovoltaic modules are growing day by day, both in standalone and grid tied mode. The large-scale installation is usually in desert and semi-arid regions because of sun's irradiance are stronger in these regions. Although Photovoltaic systems are used more in common and ease of use, they are affected by a major challenge in the large-scale installation: Dust accumulation or Soiling

PROJECT OBJECTIVES

1. Design a cleaning system which can increase the efficiency of solar panels.
2. Increase the use of solar panels.
3. Make the cleaning of solar panels simple and automated.
4. Minimize human intervention.
5. A cleaning system that does not affect the quality of the original solar panel.
6. An environmentally friendly cleaning system.

II. LITERATURE REVIEW

One of the major issues that people face with the installation and the use of solar panels is the cost that is involved in it. But the cost can be drastically reduced by increasing the efficiency of each solar panel and hence reducing the number of solar panels that needs to be installed. Using less number of solar panels in order to get the required electricity will not only be cost efficient but will also help in having a positive impact on the environment. In order to improve the efficiency of the solar panels, there are two main aspects that need to be considered, the first aspect is the amount of light that falls directly on the solar, and the second aspect is how much of this light energy is capable of using effectively in order to generate power. The issue that is faced with the use of solar panels is the dust that forms over it. Dust on the cells of the solar panels reduces the efficiency of the solar panels to a large extent especially in Saudi Arabia where dust and sand storms are very common. Hence there will be an automated system that will periodically clean the solar panels in order to make sure that they perform at peak performance level. The use of right fabrication and controllers will help in making this project possible. The Solar Photovoltaic panel cleaning technology can considerably increase the efficiency of electricity generated and also increase the durability of Solar panels. The various cleaning methods, such as electrostatic cleaning system, super hyperbolic coating methods, mechanical method, microcontroller based automatic cleaning method, self-cleaning nanodomains and various characteristics of dust particles. This project throws light on one of the various cleaning methods for solar photovoltaic panels

ARDUINO UNO

The Arduino Uno is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC to DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the ATmega8U2 programmed as a USB-to-serial converter. "Uno" means "One" in Italian. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. Arduino is an open source microcontroller which can be easily programmed, erased and reprogrammed at any instant of time. Introduced in 2005 the Arduino platform was designed to provide an inexpensive and easy way for hobbyists, students and professionals to create devices that interact with their environment using sensors and actuators. Based on simple microcontroller boards, it is an open source computing platform that is used for constructing and programming electronic devices. It is also capable of acting as a mini computer just like other microcontrollers by taking inputs and controlling the outputs for a variety of electronics devices. It is also capable of receiving and sending information over the internet with the help of various Arduino shields, which are discussed in this paper. Arduino uses a hardware known as the Arduino development board and software for developing the code known as the Arduino IDE (Integrated Development Environment). Built up with the 8-bit Atmel AVR microcontroller's that are manufactured by Atmel or a 32-bit Atmel ARM, these microcontrollers can be programmed easily using the C or C++ language in the Arduino IDE. Unlike the other microcontroller boards in India, the Arduino boards entered the electronic market only a couple of years ago, and were restricted to small scale projects only. People associated with electronics are now gradually coming up and accepting the role of Arduino for their own projects. This development board can also be used to burn (upload) a new code to the board by simply using a USB cable to upload. The Arduino IDE provides a simplified integrated platform which can run on regular personal computers and allows users to write programs for Arduino using C or C++.

ELEMENTS OF ARDUINO BOARDS

Elements of an Arduino Board can be done into two categories:

- Hardware
- Software

HARDWARE

The Arduino Development Board consists of many components that together make it work. Here are some of those main component blocks that help in its functioning.

- **Microcontroller:** This is the heart of the development board, which works as a mini computer and can receive as well as send information or command to the peripheral devices connected to it. The microcontroller used differs from board to board; it also has its own various specifications.
- **External Power Supply:** This power supply is used to power the Arduino development board with a regulated voltage ranging from 9 – 12 volts.

USB PLUG

This plug is a very important port in this board. It is used to upload (burn) a program to the microcontroller using a USB cable. It also has a regulated power of 5V which also powers the Arduino board in cases when the External Power Supply is absent:

- **Internal Programmer:** The developed software code can be uploaded to the microcontroller via USB port, without an external programmer.
- **Reset button:** This button is present on the board and can be used to resets the Arduino microcontroller.

SOFTWARE

The program code written for Arduino is known as a sketch. The software used for developing such sketches for an Arduino is commonly known as the Arduino IDE. This IDE contains the following parts in it:

- **Text editor:** This is where the simplified code can be written using a simplified version of C++ programming language.
- **Message area:** It displays error and also gives a feedback on saving and exporting the code.
- **Text:** The console displays text output by the Arduino environment including complete error messages and other information
- **Console Toolbar:** This toolbar contains various buttons like Verify, Upload, New, Open, Save and Serial Monitor. On the bottom right hand corner of the window there displays the Development Board and the Serial Port in use.

Automated Cleaning System for Solar Panels: Designing an automated cleaning system for solar panels involves the integration of robotic or mechanized devices equipped with brushes or sprayers to remove dust, debris, and other contaminants from the panel surface. This system increases panel efficiency by ensuring maximum sunlight absorption and minimizing shading effects caused by dirt buildup.

Promotion of Solar Panel Adoption: Increasing the use of solar panels requires a multi-faceted approach involving awareness campaigns, financial incentives, and policy support. By educating the public about the benefits of solar energy, offering rebates or tax credits for solar installations, and implementing regulations favoring renewable energy, the adoption of solar panels can be encouraged on a larger scale.

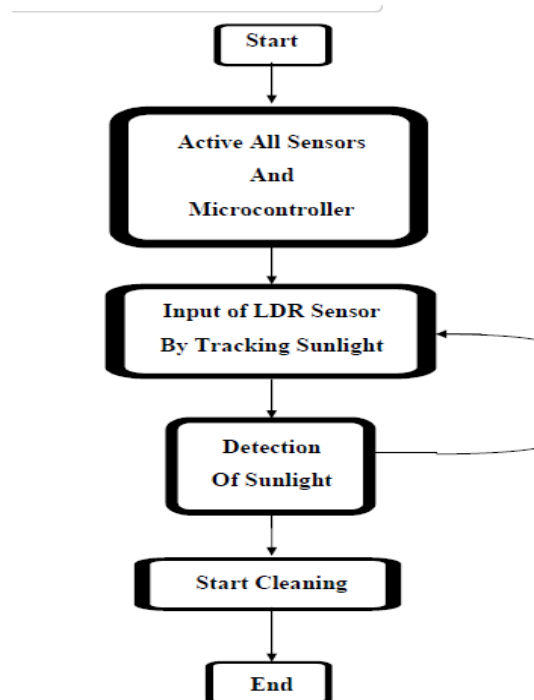
Simple and Automated Cleaning Solution: Developing a simple and automated cleaning solution for solar panels involves the use of self-cleaning coatings, hydrophobic materials, or robotic cleaning devices that operate autonomously. This eliminates the need for manual cleaning and ensures regular maintenance without human intervention, thereby optimizing panel performance.

Minimization of Human Intervention: Minimizing human intervention in the cleaning process can be achieved through the implementation of fully automated cleaning systems that operate on predetermined schedules or in response to environmental conditions. By reducing reliance on manual labor, the risk of errors and accidents associated with human intervention is minimized, leading to more efficient and reliable cleaning operations.

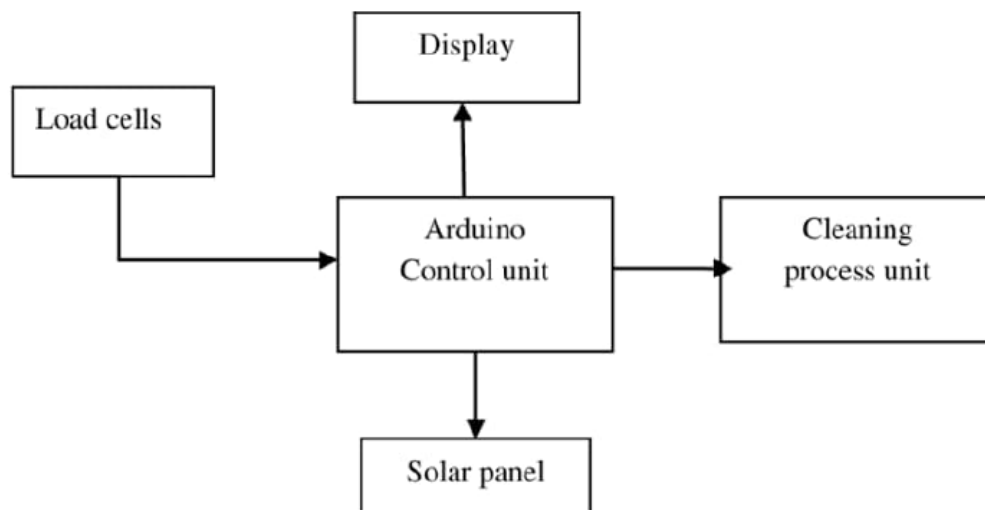
Non-Destructive Cleaning Technology: Implementing a cleaning system that does not affect the quality of the original solar panel involves the use of gentle cleaning methods such as soft brushes, air blowers, or purified water sprays. These techniques effectively remove dirt and grime without scratching or damaging the panel surface, preserving its optical properties and longevity.

Environmentally Friendly Cleaning System: Designing an environmentally friendly cleaning system entails using biodegradable cleaning agents, minimizing water consumption, and reducing energy consumption during the cleaning process. Additionally, incorporating recycling or reuse mechanisms for cleaning materials further reduces the environmental footprint of the system, ensuring sustainability throughout its lifecycle

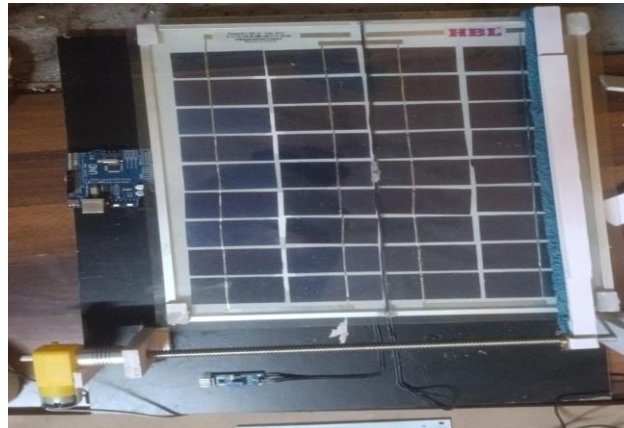
III. FLOW CHART



IV. BLOCK DIAGRAM



V. WORKING



1A solar panel is placed in the top left corner which produces dc electrical power. A microcontroller is seen just under the solar panel. A buck boost converter is shown in the right side of solar panel which takes input from the solar panel and maintains constant voltage supply. A LDR and 2 push buttons are placed in the right side of the diagram.

These above methodological circuit diagram are represented by solar panel dust cleaning mechanism. Initially the solar energy coming from sun and it can be absorbs the solar panel then by these internal circuit design mechanism these it can be stored in a battery or uses in domestic or industrial applications but here in the normal atmospheric usages of the solar panel dust cleaning the dust is frequently fan on solar panel cannot able to clean continuously. The dust is fall on panel then system output can be decrease then Arduino detects dust is on panel then it can pass information to cleaning mechanism then it clean. During this process the solar system output can be improved. Efficiency of the solar panels system can be improved this is the methodology implemented in this as in show above figure

VI. DESIGN CALCULATIONS

Wheel Radius, $r = 17.5 \text{ mm}$

No. of rotations of brush = 150 rpm

Perimeter: $2\pi r = 2 \times 3.14 \times 17.5 = 109.9 \text{ mm}$

Speed of motor = 30 rpm

$\therefore$ Distance cover in 1 minute = $30 \times 109.9 = 3297 \text{ mm/min}$

$\therefore$ Distance covered in 1 second = $3297/60 = 54 \text{ mm/sec}$

4.4.1 Torque Calculation (For Selection of Locomotion Motor) On Frame

Mass of frame, $m = 25 \text{ kg}$

Torque, $T = F \sin(\theta) \times r$ (1)

Where, $F = \text{Force} = m \times g \sin(\theta) = 20 \times 9.81 \times \sin(18) = 61 \text{ N}$

Radius, $r = 300 \text{ mm}$

$\therefore T = 61 \sin(18) \times 0.3 = 5.65 \text{ Nm}$

Taking FOS (Factor of Safety) = 3

$\therefore T = 5.65 \times 3 = 17 \text{ Nm}$

For Brush Motor:

Angular torque, $T_1 = I \alpha = MR^2 \times \alpha$ (2)

Angular acceleration $\alpha = 100 \text{ rad/sec}$

$\therefore T_1 = 12 \times 0.15^2 \times 100 = 0.405 \text{ Nm}$

Taking FOS = 3

$\therefore T_1 = 0.405 \times 3 = 1.21 \text{ Nm}$

Torque Required, $T = T_1 + \text{Frictional force}$

An additional FOS of 3 has been chosen for the Nylon Brush Motor.

$$\therefore \text{Torque} = T_1 * 3 = 1.21 * 3 = 3.63 \text{ Nm.}$$

Simulation code

```
float input_voltage = 0.0;
float temp=0.01
const int buttonPin = 3;
int MotorDir;
void setup()
{
  Serial.begin(9600);
  //IR Senfor connection
  pinMode (2, INPUT PULLUP);
  //Motor-1 connection
  pinMode (13, OUTPUT);//enable pin
  pinMode (12, OUTPUT);//input pin
  pinMode (11, OUTPUT);//input pin
  digitalWrite(13, HIGH);
  //Motor-2 connection
  pinMode (10, OUTPUT);//enable pin
  pinMode (9, OUTPUT);//input pin
  pinMode (8, OUTPUT);//input pin
  digitalWrite(10, HIGH);
  //pushbutton for direction change motors
  pinMode (buttonPin, INPUT_PULLUP);
}
void loop()
{
  int sensorval digitalRead(230);
  int reading digitalRead(buttonPin);
  int analog_value analogRead(A5);
  input_voltage (analog_value 5.0) / 1024.0;
  if (input_voltage<2)
  {
    if (MotorDir== HIGH)
    {
      if (sensorval ==LOW) {CW();Serial.println("MOTOR MOVES UP");}
    }
    else
    {
      if (sensorval== LOW) (CCW();Serial.printin ("MOTOR MOVES DOWN"))
    }
  }
}
else
{
```

```
if (MotorDir==HIGH)    (MotorDir= Low;stopMotor();)
else                    (MotorDir ==HIGH;stopMotor();)
Serial.println("MOTOR STOPS AND changes it's dizection");
delay(100);
}
}
else
{
    stopMotor();
}
}
void CW()
{
    //Motor-142 Clock wise direction rotation
    digitalWrite(12, HIGH);
    digitalWrite(11, LOW)/;
    digitalWrite(9, HIGH);
    digitalWrite(8, LOW);
    //delay14000)/// Wait for 1000 millisecond(s)
}
//Motor-1&2 Anti-clock wise direction rotation
digitalWrite(11, HIGH);
digitalWrite(12, LOW);
digitalWrite(8, HIGH);
digitalWrite(9, LOW);
//delay (4000); // Wait for 1000 millisecond(s)
}
void stopMotor()
{
    digitalWrite(11, LOW);
    digitalWrite(12, LOW);
    digitalWrite(8, LOW);
    digitalwrite (9, LOW);
```

VII. APPLICATIONS

The application of solar dust cleaning system by Arduino are given below:

- These are applicable for house hold solar panels.
- These are applicable for industrial purpose solar panels.
- These are applicable for agricultural solar panels.
- These are applicable for roof top solar panels.
- These are applicable for solar power plants.
- These are applicable for solar parking stations, etc.

ADVANTAGES

There is dirt on the surface of the solar panel, it can reduce how much light is absorbed by that area, reducing efficiency. Cleaning solar panels reduce dust, debris, and other contaminants such as leaves and bugs, which could further reduce efficiency. The advantages of solar dust cleaning system are given below:

- It improves efficiency and quality of solar panel.
- It also increases life span of solar panel.
- It required less man power.
- It requires less maintenance.
- It reduces the cost of cleaning.
- It consumes less power.
- More effective, etc.
- Less installation cost.

DISADVANTAGES

The disadvantages of solar dust cleaning system are given below:

- At initial we need to spend extra cost to install the system.
- Additional power is consumed to operate the system.
- It effects the human employment.
- We need external hardware on PV module, etc.

VIII. LIMITATIONS AND CHALLENGES

We faced many challenges in accomplishing our project. The first challenge was the machinery fabrication since we all are electrical engineering students. We faced challenges such as using mechanical software and fabricate the parts then assembly them. It is different than Arduino but we were able to do it. However, during this period of time, we faced limitation that kept us from completing our prototype in which mainly were caused by the COVID-19 pandemic. COVID-19 prevented us to merge the subsystems that were divided by our group members. In fact, it did not allow us to meet as a group and prevented some members from completing their tasks as planned

IX. FUTURE SCOPE

The device that is developed reduces the number of workers needed to clean the arrays significantly. Further Development could be done to optimize the system to be Smaller, lighter and easier to assemble in higher volumes And to become more user-friendly. The next focus will be on diversifying the robot's functionality by including auto-Inspection, communication and self-diagnostic features. The installation of a thermal camera module that will allow for inspection of the panels since the cleaning Head is in direct contact with every individual panel. Cold Spots just under the glass surface will indicate a section of Panel that remains un-cleaned and will prompt the cleaner to make another pass if needed. Solar panel energy can be used Instead of using individual battery. In future the number of solar panel are increases then simultaneously the dust in the atmosphere are also far increases then solar panel output becomes drops then only these mechanisms are most importantly in future with low cost and less human maintenance of the solar panel dust cleaning system then in future it mechanism is mostly uses this is the future scope of the automatic solar panel dust cleaning system.

- Automation: The development of fully automated solar panel cleaning machines can help to reduce the need for manual labor and make the cleaning process more efficient. This could involve the use of robotic arms or drones to clean panels on large solar farms.
- Water conservation: Currently, many solar panel cleaning machines use a lot of water to clean the panels. Future developments in technology could enable the creation of machines that use less water, making them more sustainable.
- Integration with solar panels: It may be possible to integrate cleaning mechanisms directly into the design of solar panels. For example, the use of self-cleaning coatings could reduce the need for external cleaning machines altogether.

- Smart monitoring: The use of sensors and artificial intelligence could help to monitor the performance of solar panels and identify when they need cleaning. This could help to reduce the frequency of cleaning and make the process more efficient.
- Overall, the future of solar panel cleaning machines is likely to involve a combination of automation, sustainability, and smart monitoring technologies, all aimed at improving the efficiency and effectiveness of solar panels

X. CONCLUSION

In this paper, an automatic solar panel cleaning system is proposed and built with easily available components. The proposed system is inexpensive and does not require any water to do the cleaning operation. Thus, wastage of water is avoided here. And this feature makes this system applicable in the desert areas and where no water source is available. This proposed cleaning system is based on two steps mechanism where done by wiper do the first part move in one direction which is remove dust from the surface though wiper. The second part is done by another reverse direction of wiper. This feature ensures the safety of the panel because any type of scratch is not seen during the experimental tests. Experimentally the cleaning system is capable of serving its purpose. By using this project, we can get maximum voltage or power output.

XI. REFERENCES

- [1] Website: <https://www.venturesolar.com/blog/the-importance-of-solar-energy/> [Accessed: 30- Oct.-2020].
- [2] A. Chandramouli and V. Sivachidambaranathan, "Extract Maximum Power from PV System Employing MPPT with FLC Controller," J. Elect. Eng.Automation, vol. 1, pp. 58-71, 2019.
- [3] X. Liu, S. Yue, L. Lu, and J. Li, "Investigation of the Dust Scaling Behaviour on Solar Photovoltaic Panels," J. Cleaner Production, 2021, 126391.
- [4] J. Zorrilla-Casanova et al., "Analysis of dust losses in photovoltaic modules," World Renew. Energy Cong. 2011 – Sweden, Linköping, Sweden, May 8-11, 2011, pp. 2985-92.
- [5] N. Sugiarta et al., "Preliminary design and test of a water spray solar panel cleaning system," J. Physics: Conf. Series, vol. 1450, 012108, Feb. 2020.
- [6] S. Alghamdi, A. S. Bahaj, L. S. Blunden, and Y. Wu, "Dust removal from solar PV modules by automated cleaning systems," Energies, vol. 12, pp. 1-21, 2019.
- [7] K. P. Amber et al., "A self-cleaning device for pole mounted solar photovoltaic installations," Thermal Science, vol. 23, no. 2A, pp. 739-49, 2019.
- [8] Syafaruddin, F. A. Samman, Muslimin, and S. Latief, "Design of automatic control for surface cleaning system of photovoltaic panel," ICIC Express Letters, Part B: Appl., vol. 8, no. 11, pp. 1457-64, 2017.
- [9] H. Kawamoto, "Electrostatic cleaning equipment for dust removal from soiled solar panels," J. of Electrostatics, vol. 98, pp. 11-16, 2019.
- [10] M. Mazumder et al., "Characterization of electrodynamic screen performance for dust removal from solar panels and solar hydrogen generators," IEEE Trans. Ind. Appl., vol. 49, no. 4, pp. 1793-1800, 2013.
- [11] Parrott, P. C. Zanini, A. Shehri, K. Kotsovos, and I. Gereige, "Automated, robotic dry-cleaning of solar panels in Thuwal, Saudi Arabia using a silicone rubber brush," Solar Energy, vol. 171, pp. 526-33, 2018.
- [12] S. Alagoz and Y. Apak, "Removal of spoiling materials from solar panel surfaces by applying surface acoustic waves," J. Cleaner Production, vol. 253, 2020, 119992.
- [13] G. Polizos et al., "Enhanced durability transparent superhydrophobic anti-soiling coatings for CSP applications," Proc. ASME 2014 8th Int. Conf. Energy Sustainability, Boston, Massachusetts, USA, Jun. 30- Jul. 2, 2014, pp. 1-4.
- [14] Aravind, G. Vasan, T. S. B. G. Kumar, R. N. Balaji, and G. S. Ilango, "A control strategy for an autonomous robotic vacuum cleaner for solar panels," 2014 Texas Instruments India Educators Conf., Bangalore, India, Apr. 4-5, 2014, pp. 53-61.
- [15] G. Ranganathan, "An Economical Robotic Arm for Playing Chess Using Visual Servoing," J. Innovative Image Processing, vol. 2, pp. 141-46, 2020.